

Sharon Basovich

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EDUCATION

University of Waterloo

BCS | Honours Computer Science, Co-op

Present

Waterloo, ON

INTERNSHIPS

Software Engineer | RBC | Toronto, ON

Summer 2026

- Developed and shipped features as part of a full-stack Scrum team for an internal engineering intelligence platform used by **200+ banking apps across 1,200 repositories** daily, and presented results to SVPs
- Built scheduled service-health analysis pipelines ingesting SonarQube, DAST, GitHub, and Snyk data through GraphQL/REST APIs into MongoDB, saving service owners and senior managers **12,000+ hours annually**
- Developed a React dashboard in an Nx monorepo using modular architecture to monitor cross-service health scores
- Added and maintained automated tests with Vitest and Playwright to validate app logic and reduce regressions

Quantitative Developer | Eigenvale Research | Toronto, ON

Summer 2025

- Built a **Python/NumPy/Polars backtesting platform over 3.5M daily price observations**, generating 20+ rolling equity factors and constructing weekly long/short portfolios with position, sector, and exposure constraints
- Prevented look-ahead bias using availability timestamps, point-in-time joins, and lagged execution, and built 30 tests for temporal joins, rolling windows, trade timing, and synthetic strategies with known expected returns
- Vectorized DataFrame calculations, cutting end-to-end research runtime **4.8x, from 12 minutes to 2.5 minutes**

Software Automation Engineer | Magna International | Newmarket, ON

Summer 2024

- Programmed surface-geometry, force, and leverage computations and visualizations using VBA, Excel, and CATIA V5 (CAD) to generate safety-compliance reports, **automating 50+ hours** of previously manual analysis
- Streamlined ingestion of 100+ materials records and 75+ engineering drawings into Siemens Teamcenter PLM
- Soldered wiring-harness prototypes and tested automotive hood-latch system failures in validation lab

PROJECTS

Marble Investments | \$3M AUM Investment Fund | Waterloo, ON

Present

- Built a React research graph persisting 500+ companies and 1,500+ relationships for the investment team
- Built a news pipeline monitoring **300+ tickers and 1,000–2,000 articles/day**, with LLM extraction of relevance, sentiment, affected companies, and five event categories
- Built a Python pair screener evaluating 1M+ stock pairs to shortlist roughly 50–100 candidates for analyst review

SkyCell | 1st place at Apex international hackathon | Boston, MA

Apr. 2025 – Jun. 2025

- Launched a custom 915 MHz LoRa/Meshtastic telemetry link **30 km into space** to establish a range of 70 km
- Built Python software to parse NMEA GPS data, integrate environmental sensors, and recover radio connections
- Developed a ground-station pipeline to decode telemetry and, persist flight data, and visualize it live
- Automated RTL-SDR spectrum sweeps from 30 MHz to 1.5 GHz at 2.4 MS/s, capturing I/Q data for RF analysis

Lions Robotics | Vice President & Lead Programmer | Richmond Hill, ON

Jun. 2024 – Jun. 2026

- Programmed autonomous robot PID, odometry, and Monte Carlo localization in C++, placing **10th in Canada**
- Led a 56-member organization to six World Championship qualifications while managing a \$140k annual budget

CONTESTS & AWARDS

Canadian Computing Competition Honor Roll | *Placed Top 2.5% in Senior Division*

Carnegie Mellon University picoCTF Cybersecurity Competition | *Ranked 113th out of 10,460 (Top 1%)*

VEX Robotics World Championship Judges Award | *1 of 10 Recipients Among 9,565 Teams*

10-Time Hackathon Winner | *Hack the North, General Catalyst, Ramp, etc.*

TECHNICAL SKILLS

Languages: C++, C, Python, TypeScript, Java, GraphQL, SQL, VBA

Technologies: Git, Docker, Linux, React, MongoDB, Node.js, Vitest, Playwright, PostgreSQL, OpenCV, PyTorch

Interests: Horology, Science Fiction, MMA, Poker, Skiing, Cooking, Mechanical Keyboards